DO NOW · AO1 REVISION

Energy stores and systems

Energy
Paper 1 · Combined and Separate
29 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 14 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Name the eight stores of energy.**	Chemical, electrostatic, magnetic, elastic potential, nuclear, thermal, kinetic and gravitational potential	☐
2 ★	**State the law of conservation of energy.**	Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.	☐
3 ★	**What is a closed system?**	A system where no energy enters or leaves.	☐
4 ★	**Define work done.**	The energy transferred when a force moves an object through a distance.	☐
5 ★	**State the standard unit of energy.**	Joule (J)	☐
6 ★	**Name the store of energy associated with fuels and batteries.**	The chemical energy store	☐
7	**Describe the change in the chemical energy store as fuel is burned.**	It decreases.	☐
8	**Name the store of energy associated with changes in speed.**	The kinetic energy store	☐
9	**Describe the change in the kinetic energy store as an object slows down.**	It decreases.	☐
10 ★	**Name the store of energy associated with changes in height.**	The gravitational potential energy store	☐
11	**Describe the change in the gravitational potential energy store as an object rises.**	It increases.	☐
12 ★	**Name the store of energy associated with extending and compressing objects.**	The elastic potential energy store	☐

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	QUESTION	ANSWER	✓
13	**Describe the change in the elastic potential energy store as a spring is compressed further.**	It increases.	☐
14 ★	**Name the store of energy associated with changes in temperature.**	The thermal energy store	☐
15	**Describe the change in the thermal energy store as an object gets hotter.**	It increases.	☐
16 ★	**Name the four ways energy is transferred between stores.**	Mechanically, electrically, by heating and by radiation	☐
17	**Name the energy transfer when a force moves an object.**	Mechanical transfer	☐
18	**Name the energy transfer when an object emits light or infrared.**	Transfer by radiation	☐
19	**Name the energy transfer when a hot object warms a cooler one.**	Transfer by heating	☐
20	**What is a system?**	An object or a group of objects.	☐
21	**State what happens to energy when a system changes.**	It is transferred from one store to another.	☐
22	**State the equation linking kinetic energy, mass and speed.**	kinetic energy = 0.5 × mass × speed^2 ($E_k = ½ m v^2$)	☐
23	**State the equation linking gravitational potential energy, mass, gravitational field strength and height.**	gravitational potential energy = mass × gravitational field strength × height ($E_p = m g h$)	☐
24	**State the equation linking elastic potential energy, spring constant and extension.**	elastic potential energy = 0.5 × spring constant × extension^2 ($E_e = ½ k e^2$)	☐
25 ★	**Name the two quantities that kinetic energy depends on.**	Mass and speed	☐
26	**Name the three quantities that gravitational potential energy depends on.**	Mass, gravitational field strength and height	☐

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	QUESTION	ANSWER	✓
27 ★	**A ball is thrown straight up. Describe the changes in its energy stores as it rises.**	Kinetic energy store decreases; gravitational potential energy store increases.	☐
28 ★	**An electric kettle is switched on. Describe the useful energy transfer.**	Energy is transferred electrically to the thermal energy store of the water.	☐
29 ★	**A falling apple hits the ground. Describe the energy transfer.**	Energy is transferred from the kinetic store of the apple to the thermal store of the surroundings.	☐

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